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ORIGINAL ARTICLE

Shoot-synthesized nodulation-restricting substances of wild-type soybean present in two different high-performance liquid chromatography peaks of the ethanol-soluble medium-polarity fraction

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Abstract

The nodule number of leguminous plants is controlled by “autoregulation”, a type of systemic regulation. Although many studies have attempted to identify key substances in this mechanism, none have been identified to date. Previously, we reported that the key substance(s) in this interaction are shoot-synthesized nodulation restricting substance(s) (SNRS), which are contained in the ethanol-soluble medium-polarity fraction of the shoot extract of wild-type soybean cv. Williams82. In the present study, extracts of wild-type soybean were separated by high-performance liquid chromatography (HPLC) and SNRS activity was detected in two different HPLC peaks (P-1, P-2) of the ethanol-soluble fraction. The P-1 and P-2 peaks were also found in the phloem sap of wild-type Williams82, but not in the sap of the hyper-nodulation mutant NOD1-3. A comparison of ultraviolet spectra implied that P-1 and P-2 do not contain salicylic acid, methyl jasmonate, abscisic acid, spermine or spermidine. Our results suggest that P-1 and P-2 contain two or more SNRS(s), although the purity of peaks P-1 and P-2 is still unclear.

Key words: autoregulation mechanism, bioassay, key substance, nodule, super-nodulation.

INTRODUCTION

Efficient use of dinitrogen fixation via legume–rhizobia symbioses is believed to be a core technology for sustainable agriculture. In the root nodules of legumes, host plant cells and symbiont bacteria mutually benefit from the exchange of photosynthetic and nitrogen fixation products. However, the symbiotic process is energetically expensive for the host plant, and excessive nodulation can negatively affect plant growth. Thus, a mechanism to control the number of root nodules is beneficial for the host plant.

The symbiotic process is initiated by the secretion of various substances from the roots of the host plant. These

substances, which include flavonoids (Subramanian *et al.* 2007), isoflavonoids (Phillips and Kapulnik 1995) and betaines (Phillips *et al.* 1992), induce expression of rhizobial genes involved in the rhizobia–legume symbiosis. Then, the rhizobia produce and release the nodulation signals that are recognized by the host plant. This mechanism is crucial for mutual selection between host plant and symbiont rhizobia, and is likely to play a role in the regulation of the primordial nodule number at an early stage of nodulation just after rhizobial infection. However, it is likely that other mechanisms are involved in the regulation of the number of mature root nodules. The number of primordial root nodules greatly exceeds the number of mature nodules (Francisco and Akao 1993), which suggests that the number of mature soybean nodules is regulated at a later developmental stage, rather than at an early stage of nodule formation.

For some leguminous plants, super-nodulating or hyper-nodulating mutants have been developed (Akao and Kouchi 1992; Carroll *et al.* 1985; Gremaud and Harper 1989; Jacobsen and Feenstra 1984). Such mutants have been produced via mutations in genes encoding receptor-like kinases (Krusell *et al.* 2002; Nishimura *et al.*

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2002; Searle *et al.* 2003). In cross-grafting experiments between super-nodulating or hyper-nodulating mutants and their wild-type parents, the phenotype of nodulation was determined not by root type, but rather by shoot type. The results of these experiments suggest that wild-type shoots produce certain substances that regulate nodulation (Caetano-Anolles and Gresshoff 1991; Sheng and Harper 1997). Because super-nodulating or hyper-nodulating mutants have far greater numbers of mature nodules, the systemic mechanism regulating nodule number in the mutants is assumed to be defective (Caetano-Anolles and Gresshoff 1991; Delves *et al.* 1986). This mechanism of regulation, commonly known as “autoregulation”, is believed to control root nodulation through the mediation of unidentified substance(s) produced by the wild-type shoots. Although many studies have attempted to identify the key substance in autoregulation (Bano and Harper 2002; Cho and Harper 1993; Gresshoff *et al.* 1988; Nakagawa and Kawaguchi 2006; Oka-Kira and Kawaguchi 2006; Sato *et al.* 2002; Suzuki *et al.* 2004; Terakado *et al.* 2005, 2006), none have been identified to date. Previously, we developed a unique technique to assess the effects of shoot-derived substances on root nodulation. In this technique, plantlets prepared from rooted soybean leaves are treated with various substances and their effects on nodulation are assessed (Yamaya and Arima 2004). We have used this technique to show that the hyper-nodulation trait of NOD1-3, a mutant of soybean (*Glycine max* [L.] Merr.) cv. Williams, was antagonized by a shoot extract prepared from Williams82, a wild-type derivative of cv. Williams (Yamaya and Arima 2010a). The results of that study showed that shoot-synthesized nodulation restricting substances (SNRS) were present in the shoot extracts of Williams82. It is likely that SNRS, which were present in the medium-polarity fraction of shoot extracts, are key substances in autoregulation (Yamaya and Arima 2010b).

To clarify and extend the above research, we attempted to further purify SNRS using a combination of the SNRS assessment technique and high-performance liquid chromatography (HPLC) fine fractionation of the medium-polarity fraction. We report that SNRS activity was detected in two different HPLC peaks (P-1, P-2) in extracts prepared from wild-type soybean.

MATERIALS AND METHODS

Plant material for shoot extraction

For small-scale extractions, surface-sterilized seeds of wild-type soybean *Glycine max* (L.) Merr. cv. Williams82 and its hyper-nodulation mutant NOD1-3 were sown on heat-sterilized vermiculite supplemented with a plant culture solution as described previously (Yamaya and Arima 2010a). Plants were grown in a natural light phytotron

(day/night temperatures = 25/18°C) for 3 weeks. For the large-scale extraction, 3-week-old plants (Williams82) grown in an Andosol soil field were used. The harvested shoots were frozen in liquid nitrogen and stored at –80°C until extraction.

Preparation of the medium-polarity fraction of the shoot extracts

Preparation of the soybean shoot extracts and the medium-polarity fraction of the extracts was as described previously (Yamaya and Arima 2010a, 200b). Frozen soybean shoots were ground into a powder and extracted with cold distilled water. After removing the residue by sequential filtration and centrifugation, ethanol was added to the filtrate to a final concentration of 80%, and the precipitate was removed by centrifugation (9000 g, 15 min). The supernatant was concentrated with a rotary evaporator and the high molecular weight substances were removed via centrifugal ultrafiltration (10,000 MW: Centricon Plus-20, Millipore Corporation; Billerica, MA, USA). The prepared extracts were subjected to solid-phase extraction (SPE) using a Mega BONDELUT-C18 (5 g, 20 mL) Octadecyl-bonded silica cartridge column (Varian, Palo Alto, CA, USA). The extract (20 mL) was applied to the cartridge column and was successively eluted with 30 mL distilled water and 200 mL of 80% (v/v) ethanol. The ethanol-eluted solution (medium-polarity fraction [MPF]) was concentrated to dryness under vacuum and then the dry matter was redissolved in 20 mL distilled water. The solution was filtered through a membrane filter (0.2 µm pore size) and stored at –80°C until HPLC analysis.

Phloem sap collection from Williams82 and NOD1-3 using an EDTA-method

Whole soybean plants (Williams82 and NOD1-3) grown in a field (soil type; Andosol) for 1 month were harvested and quickly washed with tap water. The plant roots were removed with scissors and the shoots were immersed in 10 mmol L⁻¹ ethylenediaminetetraacetic acid (EDTA)-Na solution (pH 6.0). The cut bases of the shoots/stems were immersed in the EDTA-Na solution for 1 h. Then, to obtain smooth cut surfaces of shoot stems, pieces of the lower stem were cut away in lengths of approximately 2 cm using a razor blade and the shoots remained immersed in the solution. The shoots were transferred into an incubation room (25°C) with saturated humidity. The bases of the shoot stems were immersed in 5 mL of the EDTA-Na solution in a glass tube for 6 h. After immersion, the solutions in the glass tubes were collected and filtered through a 0.45-µm pore size filter. The filtrate (EDTA-phloem sap) was stored at –40°C until HPLC analysis.

High-performance liquid chromatography analyses

For HPLC, we used a JASCO (Tokyo, Japan) HPLC equipped with a ultraviolet/visible (UV-VIS) multiwavelength detector (MD-915; JASCO). Analytical data were processed using a data-processing system (JASCO-PDA; DP-L1500W/DP-L910W). Extracts were separated by reverse-phase chromatography on the following ODS columns: Shodex Silica C18M-20E (20 mm $\times$ 250 mm; Showa Denko, Tokyo, Japan), Asahipak ODP-50 4E (4.6 mm $\times$ 250 mm; Showa Denko), Xbridge Shield RP18 (4.6 mm $\times$ 250 mm; Waters, Milford, MA, USA) and Xbridge Prep Shield RP18 (19 mm $\times$ 100 mm; Waters). These columns were used singly or in various combinations to detect SNRS. The mobile phases were as follows: water with 0.1% formic acid (solvent A); ethanol with 0.1% formic acid (solvent B₁); acetonitrile with 0.1% formic acid (solvent B₂); 20% ethanol with 0.1% formic acid (solvent C); 30% ethanol with 0.1% formic acid (solvent D). Solvent combinations of solvent A and B₁ (Sol-A/B₁) or solvent A and B₂ (Sol-A/B₂) were used for gradient analyses. Solvent C or D was used singly in isocratic analysis. Solvent applications to columns or to column-series were as follows: Sol-A/B₁ was used with the single Shodex Silica C18M-20E column and the tandem double-column series of Asahipak ODP-50 4E and Xbridge Shield RP18 (4.6 mm $\times$ 250 mm); Sol-A/B₂ was used with the tandem double-column series of Asahipak ODP-50 4E and Xbridge Shield RP18 (4.6 mm $\times$ 250 mm); solvent C was used with the tandem triple-column series of Shodex Silica C18M-20E and two Xbridge Prep Shield RP18 (19 mm $\times$ 100 mm) columns; solvent D was used with the single Shodex Silica C18M-20E column. The time-courses of the solvent-gradients were as follows: Pattern-1, Sol-A/B₁; 0–40 min A/B₁ = 80/20, $\rightarrow$ 41–66 min A/B₁ = 70/30, $\rightarrow$ 67–81 min A/B₁ = 40/60, $\rightarrow$ 82–103 min A/B₁ = 20/80, $\rightarrow$ 104–164 min A/B₁ = 10/90, $\rightarrow$ 165–190 min A/B₁ = 80/20; Pattern-2, Sol-A/B₂; 0–20 min A/B₂ = 80/20, $\rightarrow$ 23–43 min A/B₂ = 70/30, $\rightarrow$ 53–63 min A/B₂ = 40/60, $\rightarrow$ 73–83 min A/B₂ = 20/80, $\rightarrow$ 93–153 min A/B₂ = 10/90, $\rightarrow$ 163–183 min A/B₂ = 80/20. In all steps, linear gradient phases of 1 min (Pattern-1) or 3–10 min (Pattern-2) were used. The final step of both Pattern-1 and Pattern-2 was a conditioning phase for injection of the next sample. The flow rate was 0.8–1.0 mL min⁻¹ (ϕ 0.46 mm column and column series) or 3 mL min⁻¹ (ϕ 19–20 mm column or column series), and the sample injection volume was 10 μ L (ϕ 0.46 mm column and column series) or 100 μ L (ϕ 19–20 mm column or column series). The monitoring wavelength range was 195–500 nm for all analyses and maximum optical absorption

was recorded on chromatograms in micro absorbance units (μ AU).

In the HPLC analyses, eluates containing whole or partial peaks were divided into several fractions according to their retention time, and the SNRS activity of the respective fractions was determined. The SNRS active fractions were further separated by chromatography on different columns, and then the subfractions were evaluated for SNRS activity.

Assessment of SNRS activity

The SNRS activity of each sample solution was evaluated by measuring its inhibitory effect on the root nodulation of NOD1-3 plantlets (REL-N). These plantlets were prepared by rooting leaves excised from the first trifoliate leaf of the 3-week-old NOD1-3. To quantify the nodulation level of wild-type soybean plantlets, excised leaves of Williams82 (REL-W) was also rooted and used for assays. Details of the method have been described previously (Yamaya and Arima 2004; Yamaya and Arima 2010a).

Sample solutions and distilled water (control) were continuously introduced into the leaf apoplast of REL-N for 8 days. Only distilled water was applied to REL-W. Twenty-four hours after the solutions began to be applied, REL-Ns and REL-Ws were inoculated with a suspension of *Bradyrhizobium japonicum* (USDA110) (10⁸ cells per REL). The number of mature nodules on the roots of the RELs was counted 7 days after inoculation.

The SNRS activities of the tested solutions were evaluated based on the decrease in the number of mature root nodules of REL-N compared with that of control REL-N into which distilled water was introduced. The number of mature root nodules of REL-W was counted to determine the root-nodulation level of wild-type plantlets.

Optical absorption spectra of P-1, P-2 and six authentic compounds

We determined the optical spectra of single-peaks (P-1, P-2) with positive SNRS activity and compared them with the spectra of six authentic compounds: salicylic acid (Kanto Chemical Company, Tokyo, Japan), methyl jasmonate (Wako Pure Chemical Industries, Osaka, Japan), 3-indoleacetic acid (Kanto Chemical Company), abscisic acid (Wako Pure Chemical Industries), spermine (Kanto Chemical Company) and spermidine (Kanto Chemical Company). Optical spectra were obtained using a UV/VIS spectrophotometer (Ultrospec 3300 Pro; Amersham Pharmacia Biotech, Uppsala, Sweden), and the scanning range of wavelengths was 195–450 nm. Samples were dissolved in 90% methanol and 90% methanol was used as the control solution.

RESULTS AND DISCUSSION

Before the HPLC analyses, we re-confirmed the SNRS activity of the medium-polarity fraction of the Williams82 shoot extract using the method described previously (Yamaya and Arima 2010b).

In the first HPLC analysis, the medium-polarity fraction, which was prepared from 13 g (fresh weight) of shoots of Williams82, was charged on a Shodex C18M20E column and eluted using Pattern-1 time-course of the solvent gradient. The eluate was separated into three fractions, F1, F2 and F3 (Fig. 1a), and the SNRS activities of the fractions are shown in Fig. 1b. Compared with distilled water (control), F1, F2 and a mixture of the three fractions significantly reduced the number of nodules on the roots of REL-N. However, F3 did not significantly decrease the number of nodules. The positive SNRS activities of two fractions (F1, F2) strongly suggest that there are more than two molecular species of SNRS(s).

In the second experiment, the medium-polarity fraction was separated by HPLC in the same manner as described

in experiment 1, and then F1 and F2 were each further separated into two subfractions, the first half (F1A, F2A) and the second half (F1B, F2B) subfractions. F1A and F2B showed positive SNRS activity (data not shown).

In the third experiment, the medium-polarity fraction was separated by HPLC in the same manner as described in experiment 1, and F1A and F2B were further separated into subfractions F1A1, F1A2, F2B1 and F2B2 (Fig. 2a,b). The SNRS activities of F1A, F2B and their subfractions are shown in Fig. 2c. F1A and F2B showed significant SNRS activity. After further separation, it was found that subfractions F1A1 and F2B1 contained most of the SNRS activity.

To further isolate SNRS, the F1A1 and F2B1 subfractions were separated using another combination of columns and eluents. F1A1 was separated on a tandem double-column series consisting of an Asahipak ODP-50 4E column and an Xbridge Shield RP18 column using the Pattern-1 solvent-gradient system. The eluate was separated into subfractions S1~S5 (Fig. 3a). For F2B1, the same column series was used with the Pattern-2

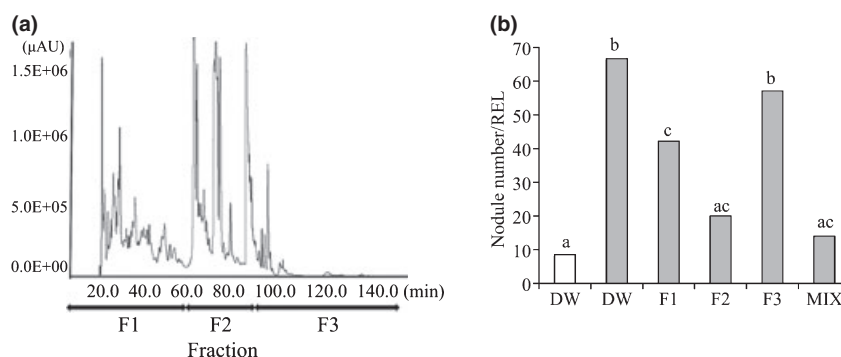


Figure 1 (a) High-performance liquid chromatography fractionation of the medium-polarity fraction of Williams82 shoot extract and (b) shoot-synthesized nodulation restricting substance activities of the subfractions. (a) The medium-polarity fraction was separated on a Shodex C18M20E column using a Pattern-1 solvent-gradient system. F1–F3, elution time bands of the subfractions of the medium-polarity fraction prepared from shoot extracts of Williams82. (b) White column, REL-W plantlet; gray columns, plantlets of REL-N; DW, distilled water; MIX, mixture of F1, F2 and F3. Different letters indicate significant differences (Tukey–Kramer test, $P < 0.05$).

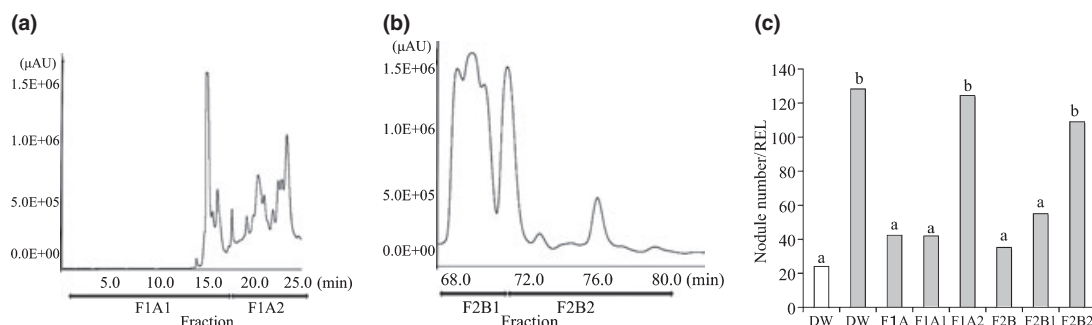


Figure 2 High-performance liquid chromatography (HPLC) fractionation of (a) F1A and (b) F2B and (c) shoot-synthesized nodulation restricting substance activity of F1A, F2B and their subfractions. (a) The HPLC conditions were the same as those described in Fig. 1. F1A1, F1A2, subfractions of F1A. (b) F2B1, F2B2, subfractions of F2B. (c) Abbreviations defined in Fig. 1.

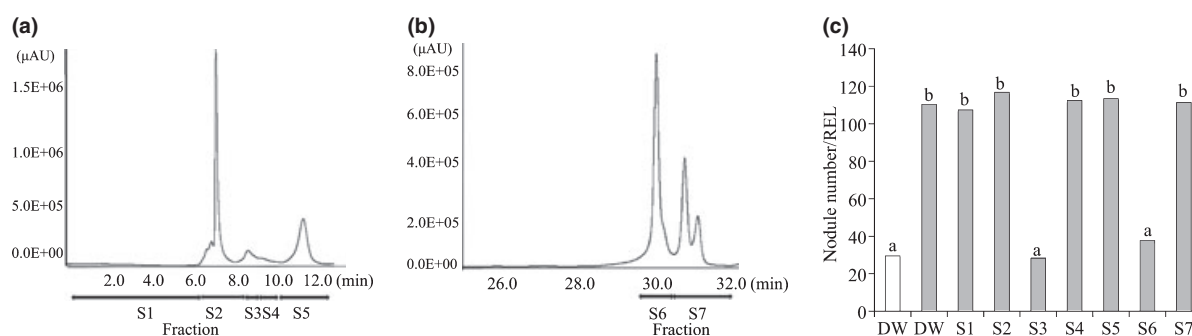


Figure 3 High-performance liquid chromatography fractionation of (a) F1A1 and (b) F2B1 and (c) shoot-synthesized nodulation restricting substance activities of their subfractions. (a) Extracts were separated on a tandem double-column (Asahipak ODP-50 4E and Xbridge Shield RP18) using a Pattern-1 solvent–gradient system. S1~S5, subfractions of F1A1. (b) Column series as described in (a) with a Pattern-2 solvent–gradient system. S6, S7, subfractions of F2B1. (c) Abbreviations defined in Fig. 1.

solvent–gradient system. The column eluate was separated into subfractions S6 and S7 (Fig. 3b). The SNRS activities of the subfractions are shown in Fig. 3c. Our results show that SNRS activity of F1A1 was contained in subfraction S3 and the activity of F2B1 was contained in S6. To further isolate the SNRS contained in S3, a sample of S3 was prepared from 100 g shoots of Williams82, and was fractionated into subfractions of S3A, S3B and S3C in the same manner as that used to fractionate F1A1 (Fig. 4a). As shown in Fig. 4b, the SNRS activity of S3 was concentrated in S3B. The dominant peak observed in S3B was designated as P-1. The dominant peak in S6 was designated as P-2.

Material(s) yielding P-1 were collected by a repeated chromatographic separation of a quantity of S3B, which was prepared from 3 kg fresh shoots of Williams82. The sample of S3B was charged on a tandem triple-column series (Shodex Silica C18M-20E and two columns of Xbridge Prep Shield RP18 [19 mm $\times$ 100 mm]) and eluted with solvent system C to collect more purified P-1 compound(s). Separation of P-1 appeared to be better

using the tandem triple-column series, and we once collected P-1 eluate for re-chromatography using the same column–eluant system. The dominant part of the materials initially mixed in the P-1 compound(s) was removed at the step of P-1 collection in the first chromatography. In the re-chromatography, P-1 showed a symmetric and smooth shape (Fig. 5a). Material(s) yielding P-2 were collected via a triple-repeat chromatographic separation of the medium-polarity fraction, which was prepared from 50 g fresh shoots of Williams82. The triple-repeat chromatographic system consisted of solvent D and a single Shodex Silica C18M-20E column. At the step of collection of P-2 eluate in the first and second chromatography, the dominant part of the materials initially mixed in the P-2 compound(s) appeared to be removed. The symmetric and smooth P-2 chromatogram detailed in Fig. 5b was given by the final triple repeat chromatography. As shown in Fig. 5c, both P-1 and P-2 showed SNRS activity. The smooth and symmetric shape of both P-1 and P-2 suggests the likelihood that the respective components are highly pure.

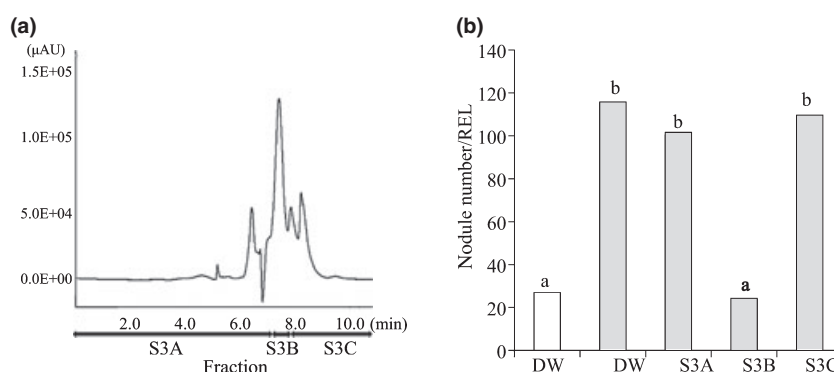


Figure 4 High-performance liquid chromatography fractionation of (a) S3 and (b) shoot-synthesized nodulation restricting substance activity of S3 subfractions. (a) The column and solvent–gradient systems were the same as those described in Fig. 3a S3A~S3C, subfraction of S3. (b) Abbreviations defined in Fig. 1.

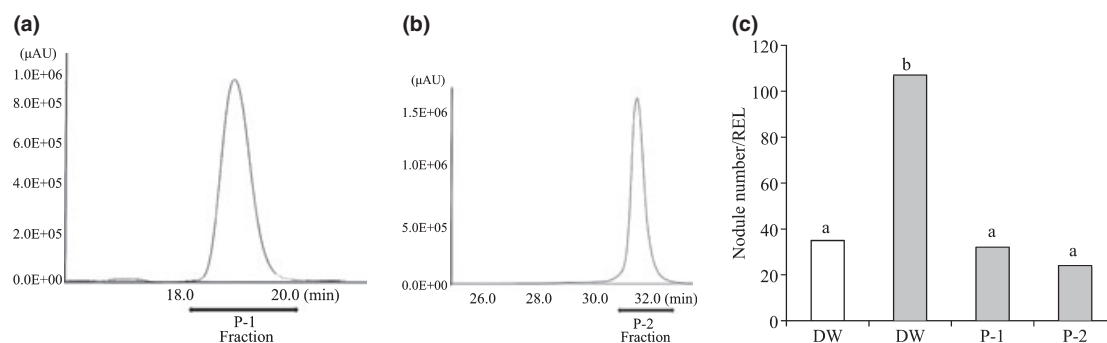


Figure 5 Collection of (a) P-1 and (b) P-2 compounds and their (c) shoot-synthesized nodulation restricting substance activity. (a) Purification of the compounds was carried out using a tandem triple-column series (Shodex Silica C18M-20E and two Xbridge Prep Shield RP18 [19 mm × 100 mm] columns) and solvent C. (b) The chromatogram of the triple-repeat chromatography for the medium-polarity fraction was obtained using a single Shodex Silica C18M-20E column and solvent D. (c) Abbreviations defined in Fig. 1.

In our SNRS assay, the tested extracts were applied to the leaflets of the plantlets and their activity was evaluated on the basis of nodulation in the roots. Therefore, it is likely that SNRSs are translocated from the shoots to the roots mainly via the phloem sieve tubes, and once they reach the roots, they control root nodulation in wild-type soybeans. Based on this idea, we chromatographically examined the phloem sap of Williams82 and NOD1-3 to determine whether or not P-1 and P-2 were present (Fig. 6). Both P-1 and P-2 were detected only in the phloem sap of Williams82. However, a peak with a longer retention time than that of P-2 was detected only in NOD1-3 phloem sap. The difference in retention time between P-2 (Williams82 phloem sap) and the peak detected only in NOD1-3 phloem sap is approximately 30 s, which is quite long enough to determine that the respective peaks are derived from different compounds, because corresponding peaks other than P-1 and P-2

showed almost no difference in the retention time between the phloem sap of Williams82 and NOD1-3. These results reveal that phloem transport of the SNRSs is found in the wild-type plants of Williams82, but not in NOD1-3.

The UV spectra of the P-1 and P-2 materials and of the six authentic compounds are shown in Fig. 7. Each spectrum was clearly different from the others, and none of the peaks within the spectra of the P-1 and P-2 materials corresponded with peaks in the authentic compounds. Although the identity of SNRS is still unclear, it is possible that SNRS is salicylic acid, methyl jasmonate, 3-indoleacetic acid, abscisic acid, spermine or spermidine. These authentic compounds have been proposed as candidate SNRS-like compounds (Bano and Harper 2002; Cho and Harper 1993; Nakagawa and Kawaguchi 2006; Sato *et al.* 2002; Suzuki *et al.* 2004; Terakado *et al.* 2005, 2006). These compounds are all distributed among a wide range of higher plants and are involved in the regulation of fundamental events in plant growth and development. However, root nodulation is a specific process that is limited to relatively few plant families, rather than a general process. Therefore, it is likely that SNRSs are compounds that are unique to nodule-forming plants.

In the present study, significant SNRS activity was detected in the compounds corresponding to P-1 and P-2, which are far distant from each other in the chromatogram of HPLC. Hence, it is evident that Williams82 shoot extracts contain two or more molecular species of SNRS in the MPF. In addition, detection of P-1 and P-2 in the phloem sap of Williams82 is the first direct evidence that indicates that SNRSs exist in phloem streaming of wild-type soybean. Further confirmation of the purity of P-1 and P-2 compounds and a large-scale purification of the SNRSs are required to conclusively identify the SNRSs. This will provide new insights into the mechanisms that restrict root-nodule formation in leguminous plants.

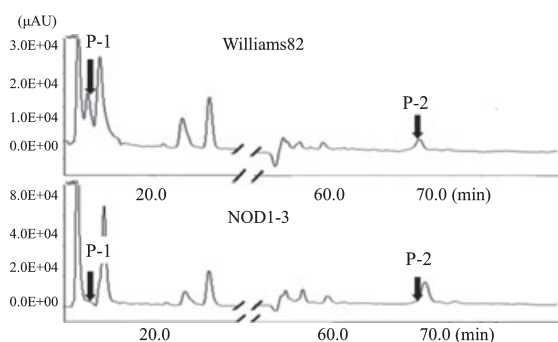


Figure 6 High-performance liquid chromatography (HPLC) chromatogram of the medium-polarity fraction prepared from phloem sap of Williams82 and NOD1-3. The HPLC conditions were the same as those described in Fig. 1. P-1 and P-2 were detected in the phloem sap of the wild-type Williams82, but not in the phloem sap of the NOD1-3 hyper-nodulation mutant.

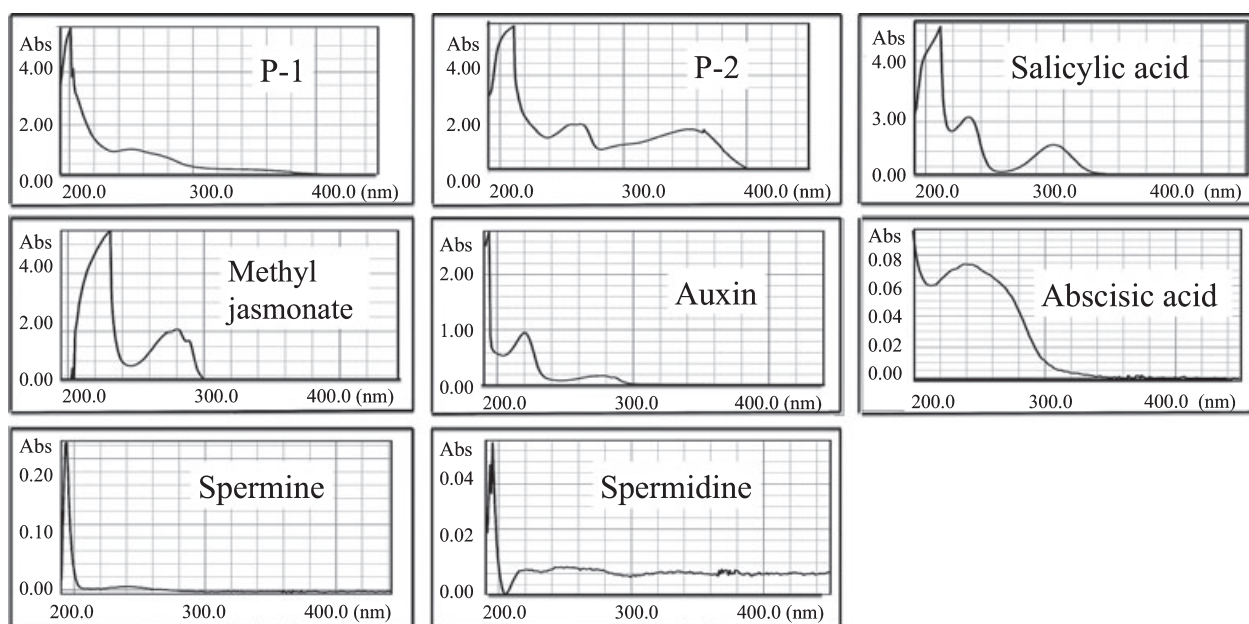


Figure 7 Comparison of the ultraviolet spectra of P-1 and P-2 with the spectra of previously proposed shoot-synthesized nodulation restricting substance candidates.

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